Weekly Lesson Plan — Week 12 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: October 26, 2026 | Course: Engineering Design & Problem Solving (InSPIRESS)

TEKS / ELPS:

TEKS: §127.406(6) implement a design process to develop a multi-system product or solution; 127.406(1) engineering disciplines and professional responsibility
ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G

Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

MONDAY OCT 26	TUESDAY OCT 27	WEDNESDAY OCT 28	THURSDAY OCT 29	FRIDAY OCT 30
Students will insert an existing model into a design, convert its bodies to components, and build a case around it with projected and offset geometry. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how engineering certifications and standards protect the public, and post photographic evidence of the day's work.	Students will model a two-part product as separate components in a Hybrid Design, activating each component before sketching in it. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how engineering certifications and standards protect the public, and post photographic evidence of the day's work.	Students will build a three-component assembly using a symmetric extrude, a circular pattern, an intersect operation and a mirror. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how engineering certifications and standards protect the public, and post photographic evidence of the day's work.	Students will manage projects, folders and versions in the Data Panel and control the view and selection tools by their official names. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how engineering certifications and standards protect the public, and post photographic evidence of the day's work.	Students will answer timed, exam-style ACU Fusion questions and then finish any missing Day 10-12 work. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how engineering certifications and standards protect the public, and post photographic evidence of the day's work.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing.*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They listen to team members' ideas, read technical directions, and write explanations of their designs. Bell work answers are written in full sentences and 2–3 students share aloud each day. Alternate-track students read, write and revise a multi-paragraph study of engineering certifications and standards.

Assessment of Content Objective: *Instrument used to determine mastery.*

Main project: the model file in each student's own cloud folder, checked against the day's build checklist and the timeline, plus the exam-style check on the ACU Fusion Hub (80% to pass, retakes allowed). Friday: the timed 12-question ACU Fusion drill. Alternate assignment: a daily photo of 3–4 written paragraphs posted to Google Classroom.

Set (Bell Work): *A motivating action that engages students in the objective. Essential Question:*

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A phone case has to protect the phone and still let every button, port and camera work. Which is harder to design, the protection or the openings? Why?	Why would a company make a dog bowl from two materials, a hard bowl and a rubber base, instead of one?	Why are screwdriver handles grooved or faceted instead of perfectly round and smooth?	(Dilemma) A teammate saved over the best version of your team's design the night before it was due. Whose fault is it, and what should an engineering team do so it never costs them a day again?	No bell work – Friday drill: 12 timed ACU Fusion questions (early dismissal).

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment.*

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Step 1: Bell ringer / essential question. Step 2: Bell work (while you sign in to Fusion + the ACU Fusion Hub): protection or openings - which is harder to design? (10 min) Step 3: Wheel: 2-3 students share their bell work (3 min) Step 4: ACU Fusion Hub - Day 10: video, build checklist, then the check (30 min) Step 5: Save (Ctrl+S) as Day 10 - Phone Case - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Certifications Day 1: what a certification is and who grants one - 3 to 4 paragraphs + a photo posted to Classroom.	Step 1: Bell ringer / essential question. Step 2: Bell work (while you sign in): why make a dog bowl from two materials? (10 min) Step 3: Wheel: 2-3 students share their bell work (3 min) Step 4: ACU Fusion Hub - Day 11: video, build checklist, then the check (30 min) Step 5: Save (Ctrl+S) as Day 11 - Dog Bowl - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Certifications Day 2: standards - why a part made in one country fits a machine made in another - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work (while you sign in): why are screwdriver handles grooved? (10 min) Step 3: Wheel: 2-3 students share their bell work (3 min) Step 4: ACU Fusion Hub - Day 12: video, build checklist, then the check (30 min) Step 5: Save (Ctrl+S) as Day 12 - Screwdriver - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Certifications Day 3: inspection - who checks the engineer's work, and what happens when nobody does - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work (while you sign in, dilemma): a teammate saved over your best version - whose fault is it? (10 min) Step 3: Wheel: 2-3 students share their bell work (3 min) Step 4: ACU Fusion Hub - Gap 1: the ten tasks in Fusion, then the check (it has real Fusion screenshots) (30 min) Step 5: Close out: every Day 10-12 file saved and named (2 min) **ALTERNATE ASSIGNMENT:** Certifications Day 4: should a certification be required to do engineering work? - 3 to 4 paragraphs + a photo posted.	Step 1: Friday drill on the ACU Fusion Hub - 12 questions, 15 minutes, no bell work today (15 min) Step 2: Wheel: 2-3 students say the question they missed and the right answer (3 min) Step 3: Catch-up: finish any of Day 10, 11, 12 or Gap 1 on the hub (25 min) Step 4: Save (Ctrl+S) - every file in Students > your name (2 min) **ALTERNATE ASSIGNMENT:** Certifications Day 5: the certification you would earn first, and your plan - a three-paragraph conclusion, plus all five photos turned in.

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, laptops, Autodesk Fusion, the "Learn Autodesk Fusion in 30 Days" videos (Days 10, 11 and 12), the ACU Fusion Hub student site (build checklists, exam-style checks with real Fusion screenshots, Friday drill), engineering notebook, alternate-assignment handout (certifications and standards).

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Rewind-and-rebuild at the teacher table for the feature that failed; the build checklist on the ACU Fusion Hub; one-on-one at the student's screen; peer re-teach inside the seating group; unlimited check retakes. Alternate track: sentence stems and a conference.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state which feature they finished and the check score they earned. Friday: the drill question they missed and the correct answer. Alternate track: students name the certification or standard they wrote about and who it protects.

Study Skills: *Skills implemented by students.*

Note-taking, technical vocabulary, following multi-step directions, self-checking against a list, timed test practice, written argument.

SPED & 504 Accommodations / Modifications: *Codes.*

Per Individualized Education Plan (IEP) / 504 plan.

ELL Accommodations: *Codes.*

Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context and rehearse with a peer before whole-class response (PN, WT). ONE-ON-ONE: individual teacher explanation on request every period; directions restated and modeled at the student desk (CD, RS, VC).

Notes: *Schedule changes this week.*

Friday Oct 30 is the end of the 2nd six weeks with early dismissal; the Friday drill replaces bell work that day. Daily routine: bell work while students sign in to Fusion and the ACU Fusion Hub, 2–3 students chosen by the wheel share, then self-paced work on the hub (video, build checklist, exam-style check). Days 10–12 and Gap 1 (Data Panel, views and selection) this week. The alternate assignment runs all five days: certifications and standards.